Clean CPU evidence: 0.5.0-registry-041f2a1

Measured source `041f2a158c3e51340b5b9e6df6886fa52584f9b5`, tree `4de2d0afb43c2127b26a5ed51d77d089016271d2`; dirty before/after: false. Library source: registry 0.5.0. One pinned CPU; f64 operations and explicitly named f32 constructors; row-major inputs.

Workload / library	Per-call p50 / p95 (us)	Batch-mean p50 / p95 (us)	Rust allocations	Peak additional requested bytes	Process peak RSS (MiB)
cholesky_factor_64_f64/faer	7.958 / 8.110	8.236 / 8.317	2	33280	4.04
cholesky_factor_64_f64/kernix	22.710 / 27.231	24.484 / 28.194	2	65536	4.07
cholesky_factor_64_f64/nalgebra	20.337 / 25.988	14.154 / 14.824	1	32768	3.84
cholesky_factor_64_f64/numpy	10.144 / 11.198	10.972 / 11.189	unavailable	unavailable	33.05
create_ones_64x64_f32/candle	0.216 / 0.231	0.230 / 0.235	7	16696	4.55
create_ones_64x64_f32/faer	0.284 / 0.292	0.172 / 0.185	1	16384	4.06
create_ones_64x64_f32/kernix	0.183 / 0.185	0.192 / 0.200	1	16384	3.89
create_ones_64x64_f32/ndarray	0.137 / 0.155	0.146 / 0.149	1	16384	3.78
create_ones_64x64_f32/numpy	1.339 / 1.592	1.325 / 1.339	unavailable	unavailable	32.69
create_zeros_64x64_f32/candle	0.152 / 0.168	0.164 / 0.171	5	16648	4.20
create_zeros_64x64_f32/faer	0.172 / 0.235	0.166 / 0.175	1	16384	4.07
create_zeros_64x64_f32/kernix	0.184 / 0.188	0.192 / 0.242	1	16384	3.82
create_zeros_64x64_f32/ndarray	0.099 / 0.100	0.094 / 0.102	1	16384	3.70
create_zeros_64x64_f32/numpy	0.383 / 0.420	0.400 / 0.497	unavailable	unavailable	32.76
matmul_128_f64/candle	70.406 / 77.392	73.624 / 77.219	9	131352	5.33
matmul_128_f64/faer	82.962 / 88.668	81.220 / 82.829	1	131072	4.59
matmul_128_f64/kernix	119.173 / 142.870	118.195 / 122.577	3	327680	4.46
matmul_128_f64/nalgebra	113.461 / 123.052	114.742 / 122.916	2	327680	4.20
matmul_128_f64/ndarray	112.665 / 126.473	118.598 / 122.508	2	327680	4.21
matmul_128_f64/numpy	74.499 / 79.894	78.699 / 83.266	unavailable	unavailable	34.09
matmul_32_f64/candle	1.496 / 1.672	1.497 / 1.512	9	8592	4.80
matmul_32_f64/faer	1.512 / 2.160	1.558 / 1.629	1	8192	4.18
matmul_32_f64/kernix	1.531 / 1.792	1.583 / 1.643	3	24576	3.98
matmul_32_f64/nalgebra	1.407 / 1.426	1.465 / 1.834	2	24576	3.79
matmul_32_f64/ndarray	1.493 / 1.669	1.444 / 1.525	2	24576	3.84
matmul_32_f64/numpy	2.620 / 3.326	2.610 / 3.154	unavailable	unavailable	32.95
matmul_sum_backward_32x16x8_f64/candle	3.698 / 4.410	4.247 / 4.456	113	27988	5.46
matmul_sum_backward_32x16x8_f64/kernix	2.951 / 3.558	2.972 / 3.106	19	18376	4.04
qr_256x64_f64/faer	426.510 / 448.319	424.901 / 455.453	7	331776	4.80
qr_256x64_f64/kernix	502.964 / 596.322	526.567 / 553.733	6	557568	4.75
qr_256x64_f64/nalgebra	380.362 / 513.470	378.185 / 423.091	4	295424	4.21
qr_256x64_f64/numpy	346.403 / 366.798	350.551 / 408.234	unavailable	unavailable	33.98
qr_96x32_f64/faer	65.855 / 76.152	70.660 / 75.438	7	67584	4.49
qr_96x32_f64/kernix	42.173 / 47.291	45.039 / 46.740	6	106752	4.16
qr_96x32_f64/nalgebra	37.531 / 42.209	37.521 / 38.272	4	57600	3.92
qr_96x32_f64/numpy	37.192 / 46.981	42.152 / 47.552	unavailable	unavailable	33.09
softmax_256x64_f64/candle	81.511 / 89.934	81.963 / 83.643	37	265200	5.64
softmax_256x64_f64/kernix	81.623 / 88.637	85.377 / 95.289	1	131072	4.20

Workload / library	Per-call p50 / p95 (us)	Batch-mean p50 / p95 (us)	Rust allocations	Peak additional requested bytes	Process peak RSS (MiB)
softmax_256x64_f64/ndarray	71.440 / 76.295	73.641 / 74.654	1	131072	3.95
softmax_256x64_f64/numpy	102.059 / 107.784	100.013 / 103.424	unavailable	unavailable	33.03

Method and limits

- Per-call quantiles use nearest ranks over 256 individual warmed calls; batch quantiles use 15 means over separate 10-ms batches. They are separate distributions. Raw samples remain in raw.json.
- Eight warmup calls follow validation. Timing includes allocation, dispatch, computation and output destruction. The first call field excludes output destruction. Inputs and library/runtime imports are prepared outside timing.
- Three extra processes record their first operation calls. These are first-use observations, not cache-flushed or device-startup latency. Timer-pair overhead is reported without subtraction; tiny calls are clock-sensitive.
- ndarray and Candle softmax are explicit stable compositions, not claims about a native softmax API.
- Constructor cases allocate owned 64x64 f32 tensors. Pre-0.6 registry baselines use full(value), explicitly marked in metadata; current workspace runs use zeros/ones. No nested literal reinterpretation is involved.
- All GEMM inputs are C-order and match elementwise. Native output layout differs by library. QR materializes thin Q and R in every implementation. Cholesky returns reusable factors (or NumPy L); no extra factor materialization is timed.
- Kernix checks finite values/symmetry. Some competitors validate a narrower contract. No finite-data precondition check was removed to improve timing. Accuracy results and budgets are retained per worker.
- The backward case uses an actual reverse-mode graph in Kernix and Candle for sum(A @ W), with A constant and W trainable. Both copy input/weight data and returned gradients; no manual derivative is used as a framework ranking.
- Allocations are measured in a separately compiled instrumented binary after warmup, outside all timing loops. Counts cover Rust global-allocator requests, not native allocator internals, stack memory or all NumPy allocations. NumPy allocation counts are unavailable, not zero.
- Requested-byte peak is additional live Rust allocation during one call, including output; it is not scratch-only memory or peak RSS. RSS is the process high-water mark including fixtures, validation and runtime, read from Linux VmHWM. Other platforms may report unavailable.
- The published cases are allocation-inclusive CPU APIs and graph workloads. Pure internal kernel-only and end-to-end application/model latency are not measured. No workload result implies a complete application frame rate.
- faer is explicitly serial; Rayon and BLAS thread counts are one. No GPU comparison. Python/NumPy timings include Python dispatch. Backend and dependency versions/configuration, governor and RAM are recorded.
- A shared laptop has uncontrolled load/frequency/cache effects. Report comparator wins and losses; no universal ranking follows. The source archive comes from committed bytes, with result directories excluded. Library dependencies are pinned by the saved driver lockfile.